

The background of the cover is a deep blue space scene. A bright, glowing light source, possibly a star or galaxy core, is visible on the right side, casting a soft, ethereal glow across the scene. On the left edge, a portion of a planet with a blue and white horizon is visible. The overall atmosphere is mysterious and cosmic.

Platonism

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Platonism is the doctrine that the spatiotemporal must be understood in terms of the non-spatiotemporal.

In this short work it will be shown that Platonism is correct.

1.0 Why we must grant that

there are non-spatiotemporal entities

Not everything that exists is in space-time. Everything that is in space-time is an **instance** of at least one **property**, and properties are not themselves in space-time. I like to play tennis. I thus have the property of liking to play tennis and am thus an instance of that property. I am obviously in space-time, but that property is not. No matter how thoroughly one searches the universe, one will never encounter the property of liking to play tennis. One will, at most, encounter its instances.

Many hold that, although property-instances exist, properties *per se* either don't exist or exist but only because they're identical with spatiotemporal entities of some kind (e.g., aggregates of their own instances, ideas in people's minds). In this paper, we'll discuss why this seemingly reasonable viewpoint cannot be sustained. We'll also see why the negation of this viewpoint is, contrary to initial appearances, utterly consonant with commonsense.

Properties themselves have properties. The property of being a square circle has the property of being uninstantiated. Properties of properties are **higher-order properties**. Properties that are not higher-order properties are **first-order** properties. The instances of higher-properties, being properties themselves, are not in space-time. The instances of first-order properties are in space-time.

Everything that exists is either a property or is in space-time. Non-spatiotemporal entities that *seem* not to be properties always turn out to be so. For example, **propositions** are not spatiotemporal and propositions do not initially seem to be identical with properties. (Propositions are sentence-meanings.) But propositions turn out to be properties, as we will soon see. **Sets** are not spatiotemporal, and sets do not initially appear to be identical with properties. But they too turn out to be properties. (But, contrary to what many a philosopher has said, properties are *not* identical with the sets that contain their own instances. The property of being a bird is not identical with the set of all birds. It's identical with a very different set, as we will later see.)

The doctrine that spatiotemporal things are instances of non-spatiotemporal things is known as "Platonism." This is because Plato was the

first person to advocate that doctrine. Sometimes Platonism is known as the *theory of universals*, and we will sometimes refer to it that way. The doctrine that properties don't exist, or do exist but only because they are identical with spatiotemporal entities, is sometimes referred to as "nominalism." But we'll usually refer to it as "anti-Platonism."

Incidentally, nominalism is so called because some anti-Platonists held that universals are identical with word. ("Nomen" being the Latin word for "name" or "word.") In Section 3.0, we'll see that this position, in addition to being false, is incoherent, the reason being that words are themselves universals.^[1]

1.1 Why we must grant that there are non-spatiotemporal entities (continued)

Given that:

(1) Sally and Fred are both intelligent

it follows that there is something that they have in common and, therefore, that

(2) There exists some property P such that Fred is an instance of P and Sally is an instance of P.

Instances of intelligence are in space-time. Einstein was in space-time; so was Keynes. But intelligence itself is not in space-time. "But mightn't the property of intelligence be nothing other than some sort of aggregate of all its instances?," one might ask. "Mightn't there be some way of spatiotemporalizing that entity?"

No. Consider the property of having a perfectly triangular shape. Nothing has this property. Are we to say that it doesn't exist? Were we to do so, we'd be saying that, since there are no instances of the property of triangularity, that property doesn't exist. Which is the same as saying there exists a

property, viz. that of triangularity, such that, because nothing has it, it doesn't exist. Which is absurd.

Also, a consequence of identifying properties with “aggregates” of their instances is that properties that clearly aren't identical with each other must be seen as being identical with each other. (In this context, the word “aggregate” can refer to any spatiotemporal entity that is composed, in *any* sense of the word “compose,” of other spatiotemporal entities.) There are no perfectly spherical objects; nor, as we just pointed out, are there any perfectly triangular objects. So, if we identify properties with aggregates of their instances, the aggregate of all the instances of the one property is identical with the aggregate of all the instances of the other, the reason being that each aggregate is a perfect nullity.

A story will clarify these points. Max is the world's greatest harpsichord player. He's also the only redhaired person who ever has, or ever will, set foot in house X. The property *red-haired person who sets foot in X* is co-extensive with the property *world's greatest harpsichord player*. Given that, for this reason, there is *no* aggregate of things having either property, it follows that no aggregate of things having the one property is different in any respect from any aggregate having the other property. This means that, anything x that is an aggregate of the one kind will be identical with anything y that is an aggregate of the other. Thus, a consequence of identifying properties with aggregates of their instances is that there cannot be distinct, uninstantiated properties. But that's absurd. The property of being a perfect husband is obviously distinct from the property of being a perfect soldier, even though neither property is instantiated.

1.2 Logic property-based, not set-based

Consider the clearly correct statement that:

(1) Squareness is a form of rectangularity.

On the face of it, (1) seems to make a statement about the property of squareness. So, if (1) 's appearance is to be trusted, then there exists a

property (viz. that of squareness) that has another property (viz. that of being a form of rectangularity). Many philosophers^[2] have tried to show that, despite its appearance, (1) doesn't really talk about properties *per se* and that it instead talks about their instances. They've proposed that the real meaning of (1) is:

(2) Everything that is square is a rectangle.

And they've proposed that other statements that, like (1), seem to refer to properties be treated similarly. (So "green is a color," they say really just means that everything green has a color.)

But (1) doesn't mean the same thing as (2). If there were exactly two square things, x and y, and x and y happened to be green, the statement:

(3) Everything that is square is green

would be true, but

(4) squareness is a form of greenness

would be false (and so would other statements of its ilk; e.g., "squareness is a color").

Of course, the statement

(5) Everything that is square is *for that very reason* a rectangle

is true, whereas

(6) Everything that is square is *for that very reason* green

is false, even if the only two square things happen to be green.

What (5) says is that:

(7) A thing's being a square is responsible for its being a rectangle,

which is obviously correct; and (6) says that:

(8) A thing's being a square is responsible for its being green,

which is obviously false. But (6) collapses into (1). (6) says that, to the extent that everything square is rectangular, it's because the property of squareness bears a certain relation (that of being a form of) to the relation of rectangularity. The relationship between a thing's being a square and its being a rectangle is, (6) rightly says, to be understood in terms of a relationship holding between squareness and rectangularity. But the relationship between a thing's being square and its being *green* is not to be understood in such terms.

A thing is a rectangle *in virtue of* its being a square, but a thing isn't green not *in virtue of* its being a square even if it happens to be a square. Whenever one encounters an expression like "in virtue of," "ipso facto," "to the extent that," or "for that very reason," one knows that one is dealing with a statement that is primarily about properties and only secondarily about objects. If I say that "copper things are ipso facto good conductors of electricity," I'm saying that the relationship between a given thing's being made of copper and its being a good conductor of electricity isn't fortuitous and is grounded in the natures of those two properties. In other words, to understand why an instance of the first property is an instance of the second, one looks to the natures of those properties. The connection goes from property instance (instance of being made of copper) to property (property of being made of copper) to property (property of being a good conductor of electricity) to property-instance (instance of being a good conductor of electricity). And the reason why "copper things are ipso facto in JMK's living room" is false is that, even though (so we will suppose for argument's sake) every copper object happens to be in my living room, the connection between x's being made of copper and x's being in my living room isn't routed in that way through those properties.

All *principles* connections are relationships between properties. This means that **nomic**^[3] connections (e.g., the connection between a thing's being made of copper and its being a good conductor of electricity) and also all logical connections (e.g., the connections between a thing's having knowledge and its having a mind) are expressed by statements whose subjects are properties. The correct statement that:

(9) For any x, the statement *x has knowledge* entails *x has a mind*

is another way of saying that

(10) something that has knowledge *for that very reason* has a mind of some kind.

And (10) is another way of saying that:

(10) the property of having knowledge is a form of the property of having a mind.

In conclusion, unless it's conceded that properties exist, there is no way to account for the fact that there are *principled* relationships of any kind. And in fact those who deny that properties exist tend *not* to grant that there exist any such relationships. Those who deny the existence of properties tend to be **empiricists**. Since empiricists cannot grant the existence of things, such as properties, that cannot be seen or otherwise known (more or less) directly on the basis of observation, they end up *denying* that anything has any bearing on anything.

1.3 Logic property-based, not set-based (continued)

Let S_1 be the set containing the smallest number greater than zero that is divisible by two, and let S_2 be Fred's favorite number, which happens to be the number two. S_1 and S_2 are the same set, and the statement:

(1) x is a member of S_1

is therefore identical with the statement

(2) x is a member of the set that contains the number two and doesn't contain anything else;

and both statements are identical with

(3) x is a member of S_2 .

Thus, anything entailed by any one of those statements is entailed by each of the others, and anything that entails any one of those statements entails each of the others. Thus, (1) entails:

(4) x is less than twenty,

and so do (2) and (3). And

(5) x is a whole number greater than one but less than three

entails (1), and it also entails each of (2) and (3).

But:

(6) x has the property of being Fred's favorite number

entails

(7) x has the property that Fred is aware of its existence,

but (1) doesn't entail this, and neither do (2) or (3). And

(8) x has the property that it is a number that is preferred by Fred to every other number that Fred has considered

entails (6), but it doesn't entail (1) or (2) or (3). It follows that properties cannot be identified with sets of their instances without mutilating the most basic principles of logic.

"But surely that's wrong," one will say. "(1) is the statement that x is a member of the set that contains the smallest number divisible by two (and nothing else)," and (2) is the statement that x is a member of the set containing Fred's favorite number (and nothing else). So I don't see how (1) and (2) could entail, and be entailed by, the same things; nor do I see how (1) and (6) could entail, or be entailed by, different things."

If the expression "the set containing Fred's favorite number" is taken in this way—in other words, if it's taken in such a way that (1) and (6) are equivalent, whereas (1) and (2) are not—then the word "set" is going proxy for the word "property." If by "the set containing Fred's favorite number" you mean, *not* "the set containing the number two," but instead "*any* set, whatever its membership, that comprises Fred's favorite number," then what

you mean by “the set containing Fred’s favorite number” is: “any set that is generated by the *property* of being Fred’s favorite number.” But in that case, when you make statements about “sets,” you’re really talking about properties.

1.4 The axiom of extensionality

It may be obvious at this point that properties aren’t identical with sets of their instances. But this now obvious point is easily extended to reveal the futility of attempts to show that properties, if existent, are spatiotemporal. But in order to make the needed extensions, we must take a very quick detour through a bit of set theory.

Two very different properties can be ***co-extensive***. Two properties are co-extensive if any instance of the one is an instance of the other. But different sets cannot have the same members. This is known as the ***axiom of extensionality***. Since different properties can generate the same set, properties aren’t identical with sets of their instances. We’ll revisit this point later. So if S_1 is a set and S_2 is a set, then S_1 is identical with S_2 if and only if nothing is a member of the one that isn’t also a member of the other. A set whose members are x , y , and z can’t fail to have those members any more than it can fail to be itself—indeed, for it to fail to have those members would be for it to fail to be itself. Thus, sets are *individuated* by their memberships. This principle is sometimes known as the axiom of extensionality.

A corollary of this axiom is that a set’s membership is frozen—frozen, indeed, in two senses. First, it’s temporally frozen. Sets don’t gain members and they don’t lose them. Second, it’s *modally* frozen. If a set’s actual members are $x_1 \dots x_n$, it couldn’t possibly exist without having exactly those members.

This may at first seem patently false. “Surely the set of people changes from year to year; people are born; people die.” This is the wrong view. The right view is that some one *property*—that of being a person—generates *different* sets at different times. So really there is no such thing as *the* set of people. There is the set of people at 3:00 P.M. April 28, 2009, the set of people at 3:01 P.M. April 28, 2009, etc.

And if *S* is the set of people—in other words, if it’s the set that contains you, me, Barack Obama, etc.—*S* couldn’t possibly exist in a universe in which one of us three didn’t exist. Of course, there could be *people* in a world where one of us didn’t exist. And, in that world, the *property* of being a person would generate *a* set. But that set wouldn’t be *S*. It would be some other set.

“But isn’t the axiom of extensionality just a matter of convention? Couldn’t we stipulate that sets *could* lose and gain members?” If we used the word “set” in this way, it would become a proxy for the word “property.” To say that the “set” of humans changes every time someone is born or dies is to say that, at any given time, this set’s membership is determined by what it is that, at that time, has the *property* of being human. So, while there’s nothing wrong with using the word “set” in this way, it isn’t an option for somebody who wishes to replace properties with sets.

Having taken care of these technical preliminaries, let us move on to matters of substance. If some piece of copper happens, at a given time, to be conducting electricity, it’s because of what is going on right then and there. The fact that there exists some other piece of copper in some remote galaxy is irrelevant. But if *S* is the set of all pieces of copper, the existence of that other piece of copper *is* relevant to *S*’s existence. Sets are individuated by their memberships. (The set consisting of Smith, Larry, and Sally ceases to exist if Smith does.) So if, for some reason, the universe hadn’t begotten that other piece of copper, then *S* wouldn’t exist. But what difference would that make to the fact that *this* piece of copper is conducting electricity? None.^[4]

Any given thing belongs to many different sets. What those sets are is only as good an indicator of how that thing is as it is an indicator of what properties that thing has. If *S* is the set of people who actually exist, then *S* doesn’t exist in a world *W* that where Dick Cheney doesn’t exist but is otherwise exactly like our world. But what difference does that make to my life-path in *W*? None. So far as you can learn anything about me from the fact that I belong to the set of humans, it’s because, in learning this, you are learning that I have the property of being human.

Everything just said about sets can be said of *any* entity that is constituted by property-instances. Suppose, for argument’s sake, that there is a “scattered object” consisting of all and only those human beings in existence. (This is a popular view among philosophers.^[5]) Just replace each occurrence of “set” in the argument just given with “scattered object consisting of all and only

human beings,” and the resulting argument shows that the property of being a person isn’t identical with that set. (Also, we’ll see in the next section that, setting aside properties that have only one instance, there are *no* spatiotemporal entities that can be identified with a given property’s instances. There are no “scattered objects.”)

2.0 Arguments against Platonism

It is seldom denied that there are instances of properties, but it has often been denied that there are nonspatiotemporal entities. But it’s hard to see (i) how there could be instances of properties without there being properties themselves, and it is also hard to see (ii) how properties could be given spatiotemporal coordinates.

Opponents of Platonism respond either by saying that (i) involves some sort of logical or linguistic trickery or by saying that properties *can* be identified with spatiotemporal entities. Let us now consider how anti-Platonists defend their view.

2.1. Argument #1 against Platonism

Not every word in a sentence corresponds to some entity in the world. If I say “for the sake of all that is good, make sure you give at least some money to charity” the word “sake” doesn’t denote anything—there aren’t such things as sakes, after all—even though, being a noun, the word “sake” has the same syntactic function as words that do have denotations.

We can develop this point by taking a brief tour through linguistic theory. It seems fairly clear that not all expressions *refer* to things. Consider words like “of,” “and,” “the,” “nonetheless.” It’s hard to believe that these words are comparable to “Smith,” “Jones,” and “cow.” The medievals referred to expressions like “Smith” and “Jones” as “categorematic” expressions, and to expressions like “of” and “the” as “syncategorematic” expressions. A syncategorematic expression is one that doesn’t refer to anything and whose linguistic function is strictly syntactical. Contemporary linguists make a similar distinction: they distinguish between “lexical” and “non-lexical”

items: the former include “Smith,” “Jones,” “red,” and “cow”; the latter include “the,” “of,” and “nonetheless.” A lexical item is one that has a referent; a non-lexical item is one that does not, and whose purpose is strictly syntactic. (In other words, the purpose of a non-lexical terms is simply to facilitate the unification of lexical items to form larger lexical items.)

Even though there are many expressions that appear to denote properties, they are really syncategorematic non-lexical expressions. Words like “red” or “the property of redness” don’t refer to anything. This is not to say that they are not meaningful. “Of” is meaningful, even though it doesn’t denote anything, and the same is true of “the property of redness.”^[6]

Argument #1 evaluated

Expressions like “red” and “the property of redness” have very different logical properties from expressions that are known to be syncategorematic. If I know that:

(1) x is red

is true, then I can logically infer

(2) There exists some color or other that x has.

This inference involves affirming the existence of the thing meant by the term “red.” Many inferences have this structure. Given:

(3) Fred is smart but Larry is not,

I can infer that

(4) there exists some property that Fred has that Larry lacks.

Words like “of” don’t behave in a comparable manner. Consider:

(5) There is a picture of Paul on the mantelpiece; nonetheless, I don’t think that the owner of the mantelpiece likes Paul, since the picture is an extremely unflattering one.

I can make many inferences from (5), but not a single one involves affirming the existence of something meant by “of” or “nonetheless.”

So it is not easy to make a case that words like “smart” and “red”—words that denote properties—lack referents and are really to be thought of on the model of purely syntactic terms like “of.”

There is another problem with Argument #1. Words are themselves universals. The word “red” isn’t identical with this or that ink mark or burst of noise. Given any inscription or burst of noise—given *any* spatiotemporal entity—that word can exist even if that spatiotemporal entity is destroyed. When you erase an inscription of the word “red,” you don’t erase *the word itself*. Inscriptions of words, and spoken utterances of them, are *instances* of words, not words *per se*. Words themselves are *properties* of such physical entities. We’ll pick this up later.

2.2 Argument #2

Although properties exist, they are identical with spatiotemporal entities. The property of being a rock is identical with the *set* of rocks. And the set of rocks is a spatiotemporal entity. That set, after all, is simply a very spread out object that consists of rocks.

Argument #2 evaluated

This answer is no good. Sets are abstract objects. They don’t have space-time coordinates. The *set* of rocks is not some big rock and it isn’t some big object consisting of rocks separated by stretches of empty space. If all the rocks in the world move to some new place, and none are destroyed, the set of rocks hasn’t changed at all: it is the same as it was before. But, under those circumstances, any object consisting of all the rocks in the world *has* changed: it has acquired a new location (or, in any case, it has undergone an internal structural change).

A set is a mathematical entity; it has no location and is defined entirely by its formal properties. *Where* its members are, and what they are doing, is irrelevant (so long as its members don’t cease to exist). But any object that is

actually *composed* of all the rocks in the world *is* affected by the movements of their members, and thus cannot be identified with sets. (Of course, it is unclear whether there is such an object—an issue we'll revisit shortly—but what is clear is that *if* there is, *then* it is affected by movements of the kind just described.) Thus, sets aren't spatiotemporal entities. Consequently, they are *themselves* Platonic entities. So one is conceding the truth of Platonism, not undermining it, by identifying properties with sets.

Aware that, for the reasons just given, sets cannot be identified with sets, philosophers of an anti-Platonist bent have sometimes chosen to identify properties with what are referred to by contemporary philosophers as *scattered objects*.

What is a scattered object? Ordinary objects are geographically and dynamically cohesive. Consider your back-pack. If you move one of the straps ten feet across the room, you have moved the whole bag across the room (unless, of course, you've damaged the bag in some way). And even though (at the microscopic level) the bag consists mostly of empty space, all of its parts occupy one, fairly compact, relatively continuous region of space-time.

Now consider all the rocks in the universe. Those rocks clearly *don't* constitute a dynamically or geographically integrated unit. Supposing that those rocks collectively constitute any object at all, the object they constitute is a "scattered object." In general, a scattered object is one that lacks the cohesiveness and dynamic integrity of ordinary objects. According to some anti-Platonists, properties are scattered objects. Is this correct?

No. There are no scattered objects. The very concept of such a thing is an incoherent one. If spatially disjoint entities are to constitute a single spatiotemporal object, their interactions must be characterized by a certain regularity and predictability. Since it cannot be known without doing empirical research whether the various rocks in the world have this property, it cannot be said whether there is some "scattered object" consisting of all the rocks in existence.

But even if scattered objects do exist, properties aren't identical with them. Let R be some scattered object composed of all the rocks in the universe. If you destroy half of the rocks in the world, you have changed the just-mentioned scattered object. But you haven't changed the conditions that a thing must satisfy to be a rock. (To be a rock, a thing must still have a certain microstructure, or whatnot.) Therefore, you haven't changed what it is

to be a rock or, consequently, the property of being a rock. So, by Leibniz's Law, that scattered object, if it even exists, cannot be identified with the property of being a rock. Given any other property, the same points *mutatis mutandis* show it not to be a scattered object.

2.3 Argument #3: The third-man argument

According to Platonism, any given spatiotemporal entity is an instance of at least one property, and for two spatiotemporal entities to resemble each other in some respect is for there to be some property of which each is an instance. Presumably, properties must resemble their instances. For example, the property of being triangular must itself be triangular. Bearing this in mind, let T be the property of being triangular, and let x be an arbitrary triangle. If Platonism is correct, it follows that there is some property T_1 such that T and x are both instances of T_1 and such that it is in virtue of their both being instances of T_1 that they are triangular. Given that, according to Platonism, properties resemble their instances, it follows that T_1 must resemble each of T and x . In other words, T_1 must be a triangle. It follows that, if Platonism is correct, there must be some property T_2 such that each of T_1 and T and x is an instance of T_2 and such that it is in virtue of their being instances of T_2 that they are triangular. And so on ad infinitum. Were they to exist, properties would be useless, since the very questions that arise in connection with their instances arise in connection with them.^[7]

Argument #3 evaluated

Consider the property of being spatiotemporal. If Platonism is right, that property is not itself spatiotemporal and, in the relevant sense of "resemble," it doesn't resemble its instances. Here is another example. To be triangular is

to have a certain shape. Only things that are in space have shapes. The property of triangularity is not in space and, therefore, doesn't have a shape and, therefore, isn't an instance of itself. TMA succeeds in showing that instantiation isn't resemblance. This means that it succeeds in proving nothing, given that it's inherent in Platonism that instantiation *not* be resemblance.

2.4 Argument #4: the argument from the causal impotence of properties

Properties lack causal powers. For this reason, they can serve no explanatory function. To explain something is to show how it came about. Being causally impotent, properties can't make anything come about. Therefore, there is no reason to posit them.

Argument #4 evaluated

Properties do indeed lack causal powers. Causes are necessarily in space and time. To be a cause is to induce change. Changes and their causes are ipso facto in space-time.

But this is no reason to think that properties are explanatory deadwood or, therefore, that there's no good reason to believe them to exist. Not all explanation is *causal* explanation. Some explanation is what we call *analytic* or *conceptual*. There are as many even numbers as there are whole numbers. (This is a theorem of set-theory.) How is this to be explained? Even though the even numbers are a subset of the whole numbers, it is possible to put the two sets of numbers into a one-one^[8] correspondence. Since, whenever two sets can be put into such a correspondence they have the same number of members, it follows that the set of even numbers has as many members as the

set of whole numbers.^[9]

This explanation is not *causal*. It explains a truth *not* by identifying its causal antecedents, but by delineating the structures of concepts. Many explanations are like this. You ask me: “Why can’t someone know something without having any form of awareness?” I answer (correctly) by saying: “because knowing something involves having a true belief, and a belief is a form of awareness.” This explanation identifies conditions that are conceptually, not causally, necessary conditions for knowledge. Like this explanation, Plato’s theory isn’t a causal theory, and it’s therefore irrelevant that properties aren’t in space-time.

2.5 Argument #5 (a variant of #4): the argument from the causal theory of knowledge (CTK)

For x to know of y, y must have causal *effects* on x. You can see the dog only because the dog has certain effects on you—light bounces off its fur and hits your eyes. You can know about Socrates only because, long ago, Socrates made various noises, which were transcribed, and copies of those transcripts made their way to. So it is only because there is a causal process linking Socrates to you that you know about him and are able to have thoughts about him.

Properties don’t have effects on anything. They don’t initiate or otherwise participate in causal processes. So we can’t know about them, even if they exist. So there is no good reason to suppose that they exist.^[10]

Argument #5 evaluated

Seeing a dog does indeed involve the dog having effects on you—it involves your eyes being disturbed by photons bouncing off of the dog. But, in and of themselves, those photonic disturbances are lifeless and

informationally empty. If they were to hit a rock, vision wouldn't result. The photons responsible for those disturbances don't contain little dog images. You have to *read such information into them*. Your mind must extract such images from them; and to do this, your mind must have some way of interpreting those disturbances. The knowledge needed to interpret these disturbances cannot come from sense-perception, since that knowledge is itself a prerequisite for sense-perception. This entails that, in at least some cases, it isn't through a causal process that the knower becomes aware of the known. It also entails that a prerequisite for the very existence of sense-perceptions, or indeed of any mental activity that results from causal exchanges with the world, is knowledge of properties. Why does it entail this? Because, were it not for a grasp of what it is for a thing to have the property of having this or that shape, or being this or that distance from one, or having this or that color, one couldn't interpret the radiation-induced disturbances of one's sensory receptors in the way needed to generate sense-perceptions.

2.6. Argument #6: Conceptualism

If they exist, properties are identical with mental representations of some kind or other—presumably, with thoughts *about* rocks or, in any case, with categories created by the mind that help it upload and process sensory information.^{[\[11\]](#)}

Argument #6 evaluated

There were rocks before there people. And, back then, rocks were similar to one another in ways that they weren't similar to bats and stars and other non-rocks. For two things to be similar to one another in a certain respect is for there to be some property that they both have. So for two rocks to resemble each other in some respect is for there to exist some property that they both have. Thus, before there were human minds, there were properties. Therefore, properties are not products of human artifice.

2.7 Argument #7: Resemblance nominalism (RN)

Let x be some red thing. x 's being red consists, not in its being related in some way or other to some non-spatiotemporal entity but in its resembling roses, spots of blood, stop-signs, and other such things. Being red is a relation, not between the spatiotemporal and the non-spatiotemporal but between the spatiotemporal and the spatiotemporal. In general, to have a property ϕ isn't to bear some relation to some other spatiotemporal entity. [\[12\]](#)

This argument evaluated: For two otherwise dissimilar square-shaped objects to resemble each other in respect of shape is for there to exist some shape, and therefore some property, that they both have. For two otherwise dissimilar red objects to resemble each other in respect to shape is for there to exist some color, and therefore some property, that they both have. There is no such thing as *mere* resemblance; there is only resemblance in this or that respect. For two things to resemble each other in a given respect is for there to exist a property that they share, and resemblance nominalism therefore collapses.

2.8 Argument #8: Conventionalism

The only non-empirical truths are linguistic conventions (e.g., "there are three feet in a yard"). Therefore, all non-conventional truths are empirical. Empirical truths register spatiotemporal facts. Conventions, whether linguistic or not, are obviously spatiotemporal entities. Therefore, all truths are spatiotemporal. [\[13\]](#)

This argument evaluated: Not all non-empirical truths are conventions. It's a non-empirical truth that there are continuous functions that cannot be differentiated at any point. But it's not a convention. *Given* the symbolic conventions that govern our usage of the relevant words ("continuous", "function," "differentiated", etc.), it *follows* that the sentence "there are continuous functions that cannot be differentiated at any point" expresses a

truth. But this means that it holds in virtue of convention *plus* some various non-conventional facts. Those facts are not empirical; they don't concern the spatiotemporal world. Therefore, the fact described by that sentence is not spatiotemporal. Further, since that fact concerns logical structures of properties (e.g., the property of being differentiable), it follows that there are properties. Since, as we just noted, that fact is non-spatiotemporal, it follows that properties are non-spatiotemporal.^[14]

3.0 Nominalism

Many medieval philosophers, and some 20th century philosophers, held that properties are identical with words and, therefore, that properties are not spatiotemporal.

But this position is incoherent, since words and linguistic expressions generally are *themselves* universals or properties. How many words are right below this sentence?

“snow”

“snow”

“snow”

Is it one or three? The answer: there are three *instances* of one *word*. To put it another way, there are three *s tokens* of one *word-type*.^[15] *Instances* of an expression are called expression-tokens and the expressions themselves are called expression-types.

No one utterance of the word “snow” is identical with that word. It's not as though the word-type “snow” will disappear when the inscriptions of it in the line above this one is destroyed. That word-type is a *property* of certain deposits of ink (e.g., the ones above) and burst of noise, and those ink-deposits and bursts of noise are therefore instances of that property. This shows that, even if the nominalist is right to replace identify the property of being a rock with the word “rock,” he has still conceded the truth of Platonism.

Plus, that property isn't identical with that word. The Spanish word for rock is “piedra.” It would be arbitrary to identify that property with the word “rock” but not with the word “piedra.” Since “rock” and “piedra” are

obviously different words, the nominalist must say that the property of being a rock is not identical with itself, which is absurd.

All of these points aside, it makes no sense to identify that property, or any other, with a word. A thing's being a rock has nothing to do with words; there were rocks before there were words.

4.0 Properties demystified

David Armstrong (1989) says that properties are “ways that things can be.” To be round is to be one way; to be square is another. Armstrong's conception of what properties are is a fruitful one; and if it is kept in mind that properties are just ways that things can be, the resistances that we have to accepting their existence crumble.

Nobody denies that there are ways things can be. To say that there are ways things can be is to say that there are possibilities as to how things can be. Not all of those possibilities are actualized. Not everything that *could be* the case is the case. Now, actualized possibilities are in space-time.^[16] But *unactualized* possibilities—possibilities that never “came true”—are surely *not* in space-time. So where are they?

One could say that they don't exist. But that would be absurd. It's one thing to say that a possibility wasn't (or won't be) *actualized*; it's a very different thing to say that the possibility *itself* doesn't exist. It's possible that you will go to law school. But (let us suppose) you're not going to go to law school (because you're going to pharmacy school instead). Does the fact that you *won't* go to law school mean that it isn't an option? Surely not. Your situation is very different from that of somebody (e.g., somebody on death-row) for whom it simply isn't an option. So the fact that a path isn't taken doesn't mean that the path doesn't *exist*; and the fact that a possibility isn't actualized doesn't mean that the possibility *itself* doesn't exist. So, given that possibilities, even unactualized ones, *do* exist, we must again ask: where are they? They're not in space-time. The *possibility* of your going to law school is not in space-time. So that possibility, being outside of space-time and thus being without location, is nowhere.

At first this may seem alarming, but it ceases to be so if we choose to see possibilities as properties and actualizations of possibilities as property-instances. It thus seems that the presumptive existence of unactualized

possibilities provides support for the theory of universals.

A story may help clarify the nature of properties. You are a meat inspector. You've just inspected all the meat in slaughterhouse X. None of it passes inspection. All of it was completely rotten. There is obviously a way something *would* have to be to be a piece of meat that passed inspection; and there is obviously a way something would have to be to be a piece of meat *in slaughterhouse X* that passed inspection. So when reporting to your supervisor, you say "none of the pieces of meat in X deserved to pass inspection," you're saying: "none of those pieces of meat was the right way." Which is the same as saying: "there exists a way that something would have to be in order to be a piece of meat in X that deserved to pass inspection; but nothing is that way." Which is the same as saying: "the property of being a piece of meat in X that deserved to pass inspection is uninstantiated."

Thus, the property of being a piece of meat in X that deserves to pass inspection isn't a giant piece of meat in the sky. It's simply a way something would have to be in order to be a piece of meat in X that passed inspection.

The same is true of the property of being a pink elephant. There are no instances of that property, since there are no pink elephants. But there is a way that something *would* have to be to be a pink elephant. I know what that way is, and so do you. That's why we both know that, by going to the zoo and spray-painting one of the elephants there, we could *make* something be that way; and that's why we also know that nothing is that way (or, in any case, that nothing we've ever heard of is that way—maybe some prankster spray painted an elephant). For something to "instantiate the property of being a pink elephant" is simply for it to be that way. So that property isn't some pink elephant in the sky. It's just a way something would have to be to be a pink elephant. And for there to be no instances of that property is simply for nothing to be that way.

Properties can be thought of as conditions. The property of being a pink elephant is a condition a thing must fulfill *in order* to be a pink elephant. Lots of unfulfilled conditions exist. Mary won't marry someone unless he fulfills the following condition: he's a genius who is always friendly and never sleeps and always dotes on her and has a billion dollars. That condition isn't fulfilled by anyone. But the condition exists. Thus, properties that aren't instantiated are just unfulfilled conditions.

So given that there are unfulfilled conditions, it follows that there are uninstantiated properties. And given that properties exist, even when they

have no instances, it follows that, when they do exist, they aren't identical with their instances. The property of being a pink elephant exists, but is currently uninstantiated. If there were to come into being a single pink elephant, which promptly died, the property wouldn't go out of existence. It would still be around. But if it were identical with instances, then it would go out of existence the moment its one instance did.

5.0 Relations and higher-order properties

We pointed out earlier that properties have properties. The property of being a bird has the property that there are more instances of it than the property of being a penguin. Actually, any statement can be interpreted as a statement about higher-order properties. "Smith has a dog" is equivalent to "the property of having a dog has the property of instantiated by Smith."

Higher-order properties are indispensable to scientific and ordinary discourse. If you say "true integrity is rare," you're saying that the property of having true integrity has the property of being seldom instantiated. If you say that ambition is more important to success than ability, you're saying that the property of being ambitious has the property of comparing favorably with the property of having ability in respect to whether, in virtue of having it, its instances are likely to succeed.

Gottlob Frege (1844–1925) discovered that statements about numbers are statements about higher-order properties. "There are zero penguins in the house" means: the property of being a penguin in the house has the property of being uninstantiated. "There are two penguins in the house" says that the property of being a pair of penguins in the house is instantiated. The number two may be defined as what all pairs of objects have in common. For x to be a pair of objects is for it to be a set S such that there are objects O_1 and O_2 , distinct from each other, that belong to S and such that anything O_2 that belongs to it is identical to either O_1 or O_2 . The number two is the property of being such a pair.

Relations may be identified with properties of ordered pairs. There is some property P such that x is taller than y exactly if the ordered pair $\langle x, y \rangle$

has P. This conception of what relations are may seem contrived, but it's an extremely useful one, since, without it, the logical forms of many inferences couldn't be identified.

6.0 Sets and propositions identical with properties

We've seen that properties are not identical with sets of their instances. The main reason for this, we saw, is properties tend to be modally elastic, whereas sets are modally frozen. In other words, a given property (e.g., the property of being a poet) can generate different instances in different worlds, whereas the *set* of poets is frozen. If Smith is a member of the set of poets in this world, and S is the set of poets in this world, then S doesn't exist in any world where Smith does not (or where Smith does exist but isn't a poet).

But sets still may be identified with properties. Suppose that Smith, Jones, and Brown are the only three poets in existence. Let S be the set of actual poets—that is, the set containing Smith, Jones, Brown, and nothing else. In that case, the statement:

(1) x is a member of S

is equivalent with

(2) x is identical with Smith or Jones or Brown

and, therefore, with

(3) x has the property of being identical with Smith or Jones or Brown.

Thus, S may be identified with the property of being identical with Smith or Jones or Brown; and membership in S may be identified with possession of that property.

Sets are non-spatiotemporal entities. Given that sets are really properties, it follows that, given only that sets are non-spatiotemporal, it doesn't follow that anything non-spatiotemporal isn't a property.

Propositions are non-spatiotemporal entities. But they too are properties.

The proposition *JM is playing tennis at time t* is true just in case certain properties are instantiated at certain times, and we may therefore identify that proposition with the set containing the relevant properties and identify that proposition's being true with the fact of all of that set's members' being instantiated. Since sets can, in the way just described, be identified with properties, it follows that propositions are properties.

Properties aren't mind-independent and neither (therefore) are propositions. But, of course, it's a popular belief, mainly among empiricists, that propositions, if existent, are human creations. Here is a reason, distinct from the one just given, to think otherwise.

There are infinitely many propositions. (Since there are infinitely many locations, there are, for any given particle x , infinitely propositions of the form *x is in place y at time t*.) We couldn't therefore have directly created every proposition there is. So, supposing that propositions are human creations, we must be only *indirectly* responsible for infinity many of them, and it must therefore be supposed that we created some finite stock of propositions that somehow brought about the rest. But in that case there are true propositions, which we'll refer to as "intermediary" propositions, that describe how the ones we directly created brought about the others. But since there are infinitely many propositions that we didn't create, there are infinitely many intermediary propositions: one for each proposition not directly created. (Suppose for argument's sake that we didn't directly create the proposition that dinosaurs wore tweed, and also suppose that $p_1...p_n$ are the propositions that we directly created. In that case, there is a true intermediary proposition to the effect that $p_1...p_n$ brought about the propositions that dinosaurs wear tweed, another intermediary proposition another to the effect that $p_1...p_n$ brought about the proposition that *either* dinosaurs wore tweed *or* $1 + 1 = 2$, and another to the effect that $p_1...p_n$ brought about the proposition that it's possible that it's the case that *either* dinosaurs wore tweed *or* $1 + 1 = 2$, and so on.) We didn't create each of these individually many intermediary propositions. (Supposing that we didn't directly create the propositions that dinosaurs wore tweed; we also didn't directly create any proposition having it as a proper part—e.g., the proposition that *either* dinosaurs wore tweed *or* $1 + 1 = 2$.) We must therefore posit a second group of intermediary propositions that will mediate between the first group and the propositions that we directly created. But everything

just said about the first group of intermediary propositions will hold of the second, and we'll be no further along. Thus, it cannot coherently be supposed that we directly create every proposition in existence. This might appear to leave open the possibility that some propositions, but not others, are human artifacts. But such a view would be arbitrary and implausible, like the view that some whole numbers are human creations while others are not. It would also be demonstrably false. We've seen that we don't directly or indirectly create the propositions describing how those that we do directly create, supposing that there are any, give rise to those we don't. Those that we don't directly create demonstrably entail those that we do. Suppose that we didn't create the proposition that dinosaurs wore tweed but that we did create the proposition that snow is white. In that case, we didn't create the proposition that *dinosaurs wore tweed and snow is white*. That proposition entails that snow is white, and thus requires its existence. Therefore the proposition that snow is white doesn't need us to create it, since its existence follows from one that we didn't create. Therefore, we *couldn't* have created the proposition that snow is white, since one can't create what already exists.

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- [1]. Russell makes this very point in *An Inquiry into Meaning and Truth*.
 - [2]. E.g. Wittgenstein (1922).
 - [3]. The word “nomic” means “having to do with natural law.”
 - [4]. See Armstrong (1993).
 - [5]. Quine (1960) held it. (See also his book *The Roots of Reference*.) In his view, the property of being water is identical with the scattered object consisting of all and only water-molecules. It’s now a popular one. I suspect that this is largely because Quine countenanced it.
 - [6]. Quine (1960, 1981) has this view. So do William of Ockham (see his book *Logical Problems*) and Bertrand Russell (1973).
 - [7]. Plato himself gives this argument in his dialogue *the Parmenides*, and so did Aristotle in the *Metaphysics*. Aristotle gave it its current name.
 - [8]. This is pronounced “one to one.”
 - [9]. See Russell (1920).
 - [10]. See Benacerraf (1965), Dretske (1982), and Fodor (1987, 1990, 1998).
 - [11]. Locke advocates this view. Many who do not explicitly advocate it are committed to it; in other words, the doctrines they do explicitly advocate are false unless conceptualism is true. Every form of empiricism presupposes the truth of conceptualism.
 - [12]. This argument is given by George Berkeley (1710).
 - [13]. Ayer (1952) and Carnap (1934) give this argument, or at least assume its cogency.
 - [14]. Quine (1966) gives a similar argument.
 - [15]. C.S. Peirce puts forth a similar argument and was the first person, so far as I know, to use the words “type” and “token” in this way.
 - [16]. In any case, anything that is in space-time is an actualized possibility. But not all actualized possibilities are in space-time. The number two, for example, is an instance of the property of being even, and is thus an actualization of the possibility of there being even numbers. But the number two is not in space-time.